Lecture 4, Jan 28-2026.

* Derivative of the Natural Logarithm function.

Thm: $$\frac{d}{dx}(\ln(x)) = \frac{1}{x}$$

Pf. Let $y = \ln(x)$ then we know

$$e^y = x$$

by implicit derivative we have:

$$\frac{d}{dx}(e^y) = \frac{d}{dx}(x)$$

$$e^y \cdot \frac{dy}{dx} = 1 \quad \text{then,}$$

$$\frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$$

Example. Differentiate $y = \ln(x^3+1)$

Sln. By the chain rule.

$$y' = \frac{1}{x^3+1} \cdot \frac{d}{dx}(x^3+1) = \frac{3x^2}{x^3+1}.$$

Rmk.

$$\frac{d}{dx}[\ln[g(x)]] = \frac{g'(x)}{g(x)}$$

Example. Find $\frac{d}{dx}(\ln(\sin x))$

Sln.

$$\frac{d}{dx}(\ln(\sin x)) = \frac{\cos x}{\sin x} = \cot x.$$

Example. Differentiate $f(x) = \sqrt{\ln x}$.

Sln.

$$f'(x) = \frac{1}{2x\sqrt{\ln x}}.$$

Example. Find $\frac{d}{dx}\left(\ln\left(\frac{x+1}{\sqrt{x-2}}\right)\right)$.

Sln.1

$$\frac{d}{dx}\left(\ln\left(\frac{x+1}{\sqrt{x-2}}\right)\right) = \frac{1}{\left(\frac{x+1}{\sqrt{x-2}}\right)} \cdot \frac{d}{dx}\left(\frac{x+1}{\sqrt{x-2}}\right)$$

$$= \frac{\sqrt{x-2}}{x+1} \cdot \frac{d}{dx}\left(\frac{x+1}{\sqrt{x-2}}\right)$$

$$\frac{d}{dx}\left(\frac{x+1}{\sqrt{x-2}}\right) = \frac{1\cdot(\sqrt{x-2}) - (x+1)\cdot\frac{1}{2\sqrt{x-2}}}{(\sqrt{x-2})^2}$$

$$= \frac{2(x-2) - (x+1)}{(x-2)2\sqrt{x-2}}.$$

$$= \frac{2x-4-x-1}{2(x-2)\sqrt{x-2}} = \frac{x-5}{2(x-2)\sqrt{x-2}}.$$

So,

$$\frac{d}{dx}\left(\ln\left(\frac{x+1}{\sqrt{x-2}}\right)\right) = \frac{\sqrt{x-2}}{x+1} \cdot \frac{x-5}{2(x-2)\sqrt{x-2}} = \frac{x-5}{2(x-2)(x+1)}.$$

Sln 2 $\ln\left(\frac{x+1}{\sqrt{x-2}}\right) = \ln(x+1) - \ln(\sqrt{x-2})$ by $\ln\left(\frac{x}{y}\right) = \ln x - \ln y$.

$= \ln(x+1) - \frac{1}{2}\ln(x-2)$ by $\ln(x^m) = m\ln(x)$.

$$\frac{d}{dx}\ln\left(\frac{x+1}{\sqrt{x-2}}\right) = \frac{d}{dx}\left[\ln(x+1) - \frac{1}{2}\ln(x-2)\right]$$

$$= \frac{d}{dx}\ln(x+1) - \frac{1}{2}\frac{d}{dx}\ln(x-2)$$

$$= \frac{1}{x+1} - \frac{1}{2}\cdot\frac{1}{x-2}$$

$$= \frac{2(x-2)-(x+1)}{2(x-2)(x+1)} = \frac{2x-4-x-1}{2(x-2)(x+1)} = \frac{x-5}{2(x-2)(x+1)}$$

Example. Find the absolute minimum of $f(x) = x^2 \ln x$.

Sln. $\text{Dom } f = (0, \infty)$. By the product rule

$$f'(x) = 2x \ln(x) + x^2 \cdot \frac{1}{x} = 2x \ln x + x = x(2 \ln x + 1)$$

$f'(x) = 0 \iff x = 0$ or $2 \ln x + 1 = 0$

$$\ln x = -\frac{1}{2}$$

$$x = e^{-1/2}.$$

So $f'(x) > 0$ then f is increasing So, $f(e^{-1/2})$ is max.

Example. Find $f'(x)$ if $f(x) = \ln|x|$

Sln. Since

$$f(x) = \begin{cases} \ln x & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}$$

it follows that

$$f'(x) = \begin{cases} \frac{1}{x} & \text{if } x > 0 \\ \frac{1}{-x} \cdot (-1) & \text{if } x < 0 \end{cases}$$

$$= \begin{cases} \frac{1}{x} & \text{if } x > 0 \\ \frac{1}{x} & \text{if } x < 0 \end{cases}$$

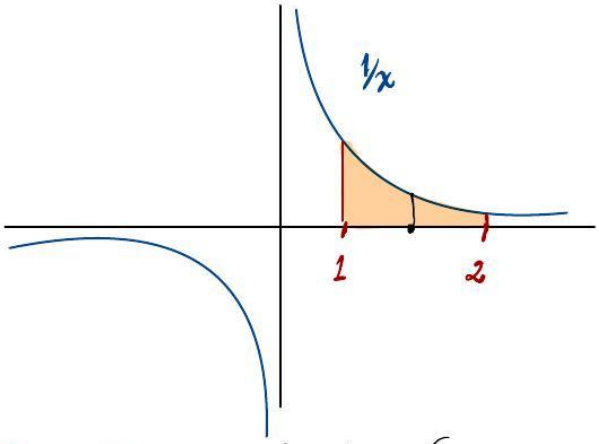
Thm. $\dfrac{d}{dx} \ln|x| = \dfrac{1}{x}.$

Thm. $\displaystyle\int \frac{1}{x}\, dx = \ln|x| + C.$

Example. Find the area under the hyperbola $xy=1$ from $x=1$ to $x=2$.

Sln. Notice that $xy=1 \iff y=\frac{1}{x}$. So,

$$A=\int_1^2 \frac{1}{x}\,dx = \ln x\Big|_1^2 = \ln(2)-\ln(1)=\ln(2)\approx 0.693.$$



Example Evaluate $\int \frac{x}{x^2+1}\,dx$.

Sln. $\int x\cdot\frac{1}{x^2+1}\,dx$. $\quad f(x)=\frac{1}{x^2+1}=h(g(x))$ $\quad h(x)=\frac{1}{x}$, $\quad g(x)=x^2+1$.

$g'(x)=2x$. let $u=x^2+1$ $\quad du=2x\,dx \iff \frac{1}{2}du=x\,dx$.

by substitution. $\quad dx=\frac{1}{2x}du$

$$\int x\cdot\frac{1}{x^2+1}\,dx=\int\frac{1}{2u}\,du=\int \cancel{x}\cdot\frac{1}{u}\cdot\frac{du}{2\cancel{x}}=\int\frac{du}{2u}$$

$$=\frac{1}{2}\int\frac{1}{u}\,du$$

$$=\frac{1}{2}\ln|u|+c.$$

$$=\frac{1}{2}\ln|x^2+1|+c.$$

$$=\frac{1}{2}\ln(x^2+1)+c \qquad \ln(x^m)=m\ln(x)$$

$$=\ln\left(\sqrt{x^2+1}\right)+c.$$

Example. Evaluate $\int_1^e \frac{\ln x}{x}\,dx$.

Sln.

$$\int \frac{\ln x}{x}\,dx = \int \frac{1}{x}\cdot \ln x\,dx.$$

$f(x) = \ln x.$

$f(x) = h(g(x))$ $\quad h(x) = x$, $\quad g(x) = \ln x$, $\quad g'(x) = \frac{1}{x}$.

let $u = \ln x \qquad du = \frac{1}{x}\cdot dx.$

$$\int \frac{\ln x}{x}\,dx = \int u\cdot du = \frac{u^2}{2} + C = \frac{1}{2}\ln^2(x) + C.$$

$$\int_1^e \frac{\ln x}{x}\,dx = \frac{1}{2}\ln^2(x)\Big|_1^e = \frac{1}{2}\left(\ln^2(e) - \ln^2(1)\right)$$

$$= \frac{1}{2}(1 - 0) = \frac{1}{2}.$$

Sln. 2.

$$\int_1^e \frac{\ln x}{x}\,dx. \qquad u = \ln(x) \qquad x = 1 \Rightarrow u = 0$$

$$x = e \Rightarrow u = 1.$$

$$= \int_0^1 u\,du = \frac{u^2}{2}\Big|_0^1 = \frac{1}{2} - 0 = \frac{1}{2}.$$

Example. Evaluate $\int \tan x\,dx$.

* General Logarithmic and Exponential functions.

$$\log_b(x) = \frac{\ln(x)}{\ln(b)}, \quad b > 1.$$

$$\frac{d}{dx}\left(\log_b(x)\right) = \frac{1}{\ln b}\cdot\frac{d}{dx}(\ln x) = \frac{1}{\ln(b)\,x}.$$

$$\frac{d}{dx}(b^x) = b^x\cdot\ln(b)$$

Also we have:

$$\int b^x dx = \frac{b^x}{\ln(b)} + C.$$

Goal: compute $\frac{d}{dx}\left(x^{\sqrt{x}}\right)$.